

Chapter 5 - The Foreign Exchange Market

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Table of contents

Introduction and Opening	2
0.1 Why This Chapter Matters	3
Chapter Roadmap	3
1 Functions of the Foreign Exchange Market	4
1.1 Function #1: Transfer of Purchasing Power	4
1.2 Function #2: Providing Credit	5
1.3 Function #3: Minimizing Foreign Exchange Risk (Hedging)	6
1.4 The Three Functions Work Together	7
1.5 The Importance of Price Discovery	7
1.6 The Global Trading Day	8
1.7 Market Participants - The Players	9
1.7.1 Bank and Nonbank Dealers	9
1.7.2 Commercial and Investment Transactors	10
1.7.3 Speculators and Arbitragers	11
1.7.4 Central Banks and Treasuries	12
1.7.5 Foreign Exchange Brokers	13
1.8 Evolution of the Market	14
1.9 The Evolution of FX Trading	15
1.9.1 1985	15
1.9.2 1990s	16
1.9.3 2010 - Today	17
1.10 The Three Components of FX Trades	19
1.10.1 Component 1: The Trade Agreement	19
1.10.2 Component 2: Messaging and Payment Instructions	19
1.10.3 Component 3: Settlement	21
1.11 FX Market Manipulation: Fixing the Fix	22
1.12 The FX Global Code of Conduct 2016	24
2 Transactions in the Foreign Exchange Market	25
2.1 Spot Transactions	25
2.2 Forward Transactions	27

2.3	Swap Transactions	29
2.3.1	Spot Against Forward	29
2.3.2	Forward-Forward Swaps	30
2.3.3	Nondeliverable Forwards (NDFs)	31
2.4	Size of the Foreign Exchange Market	34
2.5	Geographical Distribution	35
2.6	Currency Composition	36
3	Foreign Exchange Rates and Quotations	37
3.1	Currency Symbols	38
3.2	Exchange Rate Quotes	39
3.3	Market Conventions	41
3.3.1	European Terms	41
3.3.2	American Terms	42
3.3.3	Currency Nicknames	43
3.3.4	Direct and Indirect Quotations	43
3.3.5	Bid and Ask Rates	45
3.4	Cross Rates	48
3.5	Intermarket Arbitrage	52
3.6	Forward Quotations	57
	Conclusion and Synthesis	63
3.1	Key Takeaways	63
3.2	Connecting to Broader Course Themes	64
3.3	Practical Applications	64
3.4	The Mini-Case: Iceland	65
3.5	Preparing for Success	65
3.6	Common Pitfalls to Avoid	65
3.7	Questions for Reflection	66
3.8	Looking Ahead	66
3.9	Final Thoughts	67
	Questions and Discussion	67

Introduction and Opening

This chapter may be the largest, most active, and most globally integrated financial market in the world—the foreign exchange market. With daily trading volumes exceeding \$6.6 trillion, this market dwarfs even the world’s stock markets.

Opening Quote: As we begin, consider this observation from John Maynard Keynes:

“The best way to destroy the capitalist system is to debauch the currency. By a continuing process of inflation, governments can confiscate, secretly and unobserved, an important part of the wealth of their citizens.”

This quote reminds us that currency markets aren’t just about converting dollars to euros for your vacation—they’re fundamental to the functioning of the global economy and the wealth of nations.

Opening Question to Class: “How many of you have ever exchanged currency, either at an airport kiosk, through an ATM abroad, or via an online platform?”

Every time you make that exchange, you’re participating in the foreign exchange market—the same market where central banks intervene to protect their currencies, where multinational corporations hedge billions in currency exposure, and where speculators try to profit from currency movements.

0.1 Why This Chapter Matters

Understanding the foreign exchange market is crucial for several reasons:

1. **For International Business:** Any company engaged in cross-border trade must understand how to navigate FX markets, manage currency risk, and make informed decisions about when and how to exchange currencies.
2. **For Financial Management:** Corporate treasurers at multinational firms must understand FX markets to hedge currency exposures, optimize cash management across currencies, and manage the firm’s overall currency risk profile.
3. **For Investment Analysis:** Currency movements can dramatically impact investment returns. A 10% gain in a foreign stock market means nothing if the foreign currency falls 15% against your home currency.
4. **For Economic Understanding:** Exchange rates impact trade balances, inflation, interest rates, and economic growth. Understanding FX markets is essential for understanding the global economy.

Chapter Roadmap

Today we’ll cover four major learning objectives:

1. **Functions of the Foreign Exchange Market** - What purposes does this market serve in the global economy? We’ll explore the three core functions: transferring purchasing power, providing credit, and minimizing FX risk.
2. **Structure of the Foreign Exchange Market** - How is this massive, 24-hour global market organized? Who are the players? How has technology transformed trading? We’ll trace the evolution from telephone trading to today’s electronic platforms.
3. **Transactions in the Foreign Exchange Market** - What types of transactions occur? We’ll examine spot transactions, forward contracts, swaps, and more exotic instruments like non-deliverable forwards (NDFs).
4. **Foreign Exchange Rates and Quotations** - How are exchange rates quoted? What do terms like “bid,” “ask,” “direct,” “indirect,” “American terms,” and “European terms” mean? How do we calculate cross rates and identify arbitrage opportunities?

By the end of today's session, you should be able to read and interpret currency quotations, understand how different market participants interact, calculate cross rates, identify arbitrage opportunities, and appreciate how this market has evolved from voice brokers to sophisticated electronic trading platforms.

1 Functions of the Foreign Exchange Market

Let me start with a fundamental definition that we'll build upon throughout this chapter:

Foreign Exchange Market Definition: The foreign exchange market provides the physical and institutional structure through which the money of one country is exchanged for that of another country, the rate of exchange between currencies is determined, and foreign exchange transactions are physically completed.

Think of this as the “plumbing” of the global economy. Just as your home needs plumbing to function, the global economy needs the foreign exchange market to function.

Key Quote from Federal Reserve: Simon Potter, Executive Vice President of the Markets Group of the Federal Reserve Bank of New York, said at the 2015 FX Week Conference:

“The foreign exchange market is perhaps the largest, most globally integrated, and most active financial market in the world. The transactions that take place there are the lifeblood of a global economy that comprises many different national currencies. By enabling the transfer of funds and purchasing power from one currency to another, the foreign exchange market offers an important means for price discovery that facilitates international trade and investment activity.”

Now, let's examine the three fundamental functions of this market:

1.1 Function #1: Transfer of Purchasing Power

The Core Problem: International trade and capital transactions normally involve parties living in countries with different currencies. Usually each party wants to deal in its own currency, but the trade or capital transaction can be invoiced in only ONE currency. Hence, one party must deal in a foreign currency.

Example - A Simple Import Transaction:

Imagine you're a U.S. retailer importing wine from France:

- You (U.S. retailer) think in dollars
- French vineyard thinks in euros
- The invoice will be in either dollars OR euros, not both
- Someone must exchange currencies

If the invoice is in euros, YOU must:

1. Buy euros with your dollars
2. Pay the French vineyard in euros
3. The French vineyard keeps euros (their home currency)

If the invoice is in dollars, the French vineyard must:

1. Receive your dollar payment
2. Sell those dollars for euros
3. Keep euros for their operations

The FX Market's Role: The foreign exchange market enables this conversion—transferring your purchasing power in dollars into the French vineyard's purchasing power in euros.

Scale of This Function: With over \$25 trillion in global merchandise trade annually, and even more in services and capital flows, this purchasing power transfer function is enormous. Every single international transaction requires it.

1.2 Function #2: Providing Credit

Because the movement of goods between countries takes time, inventory in transit must be financed. The foreign exchange market provides a source of credit, in addition to specialized instruments like:

- **Bankers' acceptances:** A bank's promise to pay, used to finance international trade
- **Letters of credit:** A bank's conditional commitment to pay on behalf of its customer

Why Credit Matters in FX Markets:

Scenario: A U.S. company imports machinery from Germany.

- Machinery is shipped from Hamburg on January 1
- Machinery arrives in Los Angeles on February 15
- U.S. company has 30 days to pay after arrival (payment due March 15)

The Timeline Creates Credit Needs:

- German exporter has already paid to manufacture the machinery
- Machine is in transit for 45 days
- U.S. importer has an additional 30 days to pay
- Total time: 75 days of financing needed

The FX market helps provide this credit in several ways:

1. **Forward Contracts:** The U.S. importer can lock in an exchange rate TODAY for a payment 75 days from now, essentially obtaining currency on credit
2. **Currency Swaps:** A sophisticated way to borrow in one currency while simultaneously lending in another
3. **Working with Banks:** Banks provide trade finance, essentially extending credit in foreign currency

Important Point: The FX market doesn't just facilitate immediate spot exchanges—it provides the credit mechanisms that make international trade's time lags manageable.

1.3 Function #3: Minimizing Foreign Exchange Risk (Hedging)

The Risk Problem: When you have obligations or receivables in foreign currency, you face foreign exchange risk—the risk that exchange rates will move against you.

Example of FX Risk:

Imagine you're a U.S. exporter selling manufacturing equipment to Japan:

- Contract signed: January 1
- Price: ¥100 million
- Payment due: July 1 (6 months later)
- Today's exchange rate: ¥100 = \$1.00
- Expected dollar revenue: \$1,000,000

What Could Go Wrong?

By July 1, the yen could weaken:

- New exchange rate: ¥120 = \$1.00
- Your ¥100 million now equals: \$833,333
- You've "lost" \$166,667 just from currency movement!

The Hedging Solution:

On January 1, you could enter a forward contract:

- Agree TODAY to sell ¥100 million
- For delivery in 6 months (July 1)
- At a rate of ¥101 = \$1.00 (forward rate)
- Guaranteed dollar revenue: \$990,099

You've given up \$10,000 of potential profit, but you've eliminated \$166,667 of risk. This is hedging—using the foreign exchange market to minimize exposure to exchange rate changes.

Types of Hedging Instruments:

1. **Forward Contracts:** Agreement to exchange currencies at a future date at a predetermined rate
2. **Futures Contracts:** Standardized forwards traded on exchanges
3. **Options:** Right (but not obligation) to exchange at a specified rate
4. **Swaps:** Exchange one currency for another temporarily

Critical Insight: The foreign exchange market provides these "hedging facilities" for transferring foreign exchange risk from one party to another. Someone willing to bear risk (speculators) can take the other side of hedging transactions from those seeking to avoid risk (hedgers).

1.4 The Three Functions Work Together

Think of a major multinational corporation like Apple:

- **Transfer Function:** Apple sells iPhones in Europe, must convert euro revenues to dollars
- **Credit Function:** Apple uses currency swaps to finance operations in different currencies
- **Hedging Function:** Apple hedges billions of dollars of foreign currency exposure using forwards, options, and other derivatives

These three functions working simultaneously make global trade and investment possible. Without an efficient foreign exchange market providing all three functions, international business as we know it could not exist.

1.5 The Importance of Price Discovery

Beyond these three explicit functions, the FX market performs an implicit fourth function: **price discovery**.

With trillions of dollars traded daily, the FX market aggregates information from millions of participants to determine what currencies are worth relative to each other. This price discovery:

- Reflects economic fundamentals (interest rates, inflation, growth)
- Incorporates expectations about future policies
- Responds instantly to news and events
- Provides transparent pricing to all participants

Bottom Line: The foreign exchange market is the mechanism by which the global economy reconciles the fact that different countries use different currencies. It transfers purchasing power, provides credit, enables hedging, and discovers prices—all functions essential to international commerce and investment.

—## Structure of the Foreign Exchange Market

The foreign exchange market has evolved dramatically over time—from money changers in the streets of Florence and Venice, to trading rooms in London and New York, to today's world where a trader with a handheld computer anywhere in the world can execute a multi-million dollar trade in seconds. Yet despite this technological transformation, the market remains based on the same fundamentals: supply and demand, market information and expectations, and negotiating strength.

Key Characteristics of the FX Market:

- **No central location:** Unlike the New York Stock Exchange, there's no single building or trading floor
- **24-hour trading:** The market never sleeps—currencies trade somewhere every hour of every business day
- **Decentralized:** Trading occurs simultaneously in multiple locations worldwide
- **Electronic:** Increasingly dominated by computer trading and algorithms
- **Informal yet regulated:** No single global regulator, but individual countries regulate their banks' participation

Let's explore how this unique market is structured.

1.6 The Global Trading Day

The Sun Never Sets on Currency Trading

The foreign exchange market operates 24 hours a day, but people—traders—do eventually sleep. This translates into four major trading sessions per day, following the sun around the globe:

The Daily Cycle:

1. **Asian Session:** Trading day begins each morning in Sydney and Tokyo
2. **Asian/Middle East:** Moves to Hong Kong, Singapore, and Middle Eastern centers
3. **European Session:** Passes to Frankfurt, Zurich, and London
4. **American Session:** Makes the final jump across the Atlantic to New York and Chicago
5. **West Coast:** Ends in San Francisco and Los Angeles

Important Timing Note: Many large international banks operate foreign exchange trading rooms in each major geographic trading center to serve both their customers and conduct proprietary trading on a 24-hour basis.

Peak Trading Hours:

Although global currency trading is indeed a 24-hour activity, not all hours are equally busy:

- **Highest Volume:** When London and New York overlap (roughly 8am-12pm EST)
 - These two centers together account for over 70% of global FX trading
 - Greatest depth and liquidity during this overlap
- **Growing Asian Influence:** Hong Kong, Tokyo, and Singapore are increasing their market share
 - Chinese renminbi growth is significant
 - Shanghai's influence is rising as China's economy expands

The Twilight Hour Problem:

Your textbook describes an interesting phenomenon called the “twilight hour”—the period between roughly 5-7 PM New York time when:

- New York is closing, traders heading home
- Hong Kong/Sydney traders are just starting their day
- Market liquidity is thin
- Strange price movements can occur

Real Examples:

- **January 3, 2019:** Japanese yen suddenly surged 3% against the dollar during the twilight hour after Apple released weak China sales data
- **October 7, 2016:** British pound fell 6% to a 31-year low during twilight trading after speculation about a “hard Brexit”

Lesson: Releasing important information during periods of thin trading can cause disproportionate market reactions. Both governments and corporations should be aware of global trading patterns when making announcements.

1.7 Market Participants - The Players

The foreign exchange market serves a diverse group of participants. We can simplistically divide them into two major groups:

1. **Liquidity Seekers:** Those trading currency for commercial purposes (to facilitate trade/investment)
2. **Profit Seekers:** Those trading for profit (speculators, arbitrageurs)

Historical Context: The FX market began as a market for liquidity—facilitating the exchange of currency for commerce. But the exceptional growth in recent decades has been largely driven by profit seekers.

Important Observation: The profit seekers are typically much better informed about the market. They study it full-time, use sophisticated models, and have access to the best information. As might be predicted, profit seekers generally profit from liquidity seekers.

Now let's examine five broad categories of participants:

1.7.1 Bank and Nonbank Dealers

Primary Role: Market makers who stand ready to buy and sell currencies

How They Profit: Bank and nonbank dealers profit from buying foreign exchange at a **bid price** and reselling it at a slightly higher **ask price** (also called “offer”). The difference between bid and ask is the **spread**—this is their profit margin.

Example:

EUR/USD Bid: 1.1270

EUR/USD Ask: 1.1274

Spread: 0.0004 or 4 "pips"

If a bank buys €10 million at the bid (paying \$11,270,000) and immediately sells it at the ask (receiving \$11,274,000), they make \$4,000 profit in seconds.

Market Making:

- Large international banks function as “market makers”
- They stand willing at all times to buy and sell currencies in which they specialize
- They maintain an “inventory” position in those currencies
- They trade with other banks worldwide to maintain inventories within limits set by bank policy

Trading as a Profit Center:

- Foreign exchange departments of many banks operate as profit centers
- Individual dealers are compensated based on profitability of their trading
- Major banks earn 10-20% of annual net income from currency trading
- Traders can earn substantial bonuses based on their trading profits

Competition:

Competition among dealers worldwide narrows the spread between bids and asks, making the foreign exchange market very efficient. For major currencies like EUR/USD, spreads can be just a few hundredths of a percent—far narrower than retail exchange rates you’d see at airports.

Size Matters:

- Small to medium-sized banks participate but are not typically market makers
- They buy from and sell to larger institutions to offset retail transactions
- They don’t maintain large inventory positions

1.7.2 Commercial and Investment Transactors

Who They Are:

- Importers and exporters
- International portfolio investors
- Multinational corporations
- Tourists
- Anyone conducting international commerce or investment

Their Motivation: Their use of the foreign exchange market is **necessary but incidental** to their underlying commercial or investment purpose. They’re the quintessential “liquidity seekers.”

Example - An Importer:

A U.S. company importing wine from France needs euros to pay the French supplier. The company:

- Isn’t trying to profit from currency trading
- Simply needs to convert dollars to euros
- Wants to complete this transaction efficiently and at a fair price
- May want to hedge the exchange rate risk

Hedging Behavior:

Some of these participants use the market not just to exchange currency, but to **hedge foreign exchange risk**:

- Importers can lock in future rates using forward contracts
- Exporters can protect themselves against currency depreciation
- Portfolio investors can hedge their foreign stock/bond holdings

Important Insight: These participants provide a steady flow of business to the market but generally aren’t as sophisticated as the dealers and speculators. They often end up paying the bid-ask spread, which represents a transfer of wealth to the dealers.

1.7.3 Speculators and Arbitragers

The True Profit Seekers

These participants seek to profit from trading within the market itself. They operate in their own interest, without obligation to serve clients or ensure a continuous market.

Speculators:

- Seek profit from **exchange rate changes**
- Take positions based on their views about future currency movements
- Bear risk in hopes of profit

Example:

A speculator believes the euro will strengthen against the dollar:

- Today: $\text{EUR/USD} = 1.1000$
- Speculator buys €10 million (paying \$11 million)
- One month later: $\text{EUR/USD} = 1.1500$
- Speculator sells €10 million (receiving \$11.5 million)
- Profit: \$500,000

Arbitragers:

- Try to profit from **simultaneous exchange rate differences** in different markets
- Seek “risk-free” profits by exploiting temporary price discrepancies
- Require very fast execution and low transaction costs

Example of Triangular Arbitrage:

Bank A: $\text{EUR/USD} = 1.1000$

Bank B: $\text{GBP/USD} = 1.3000$

Bank C: $\text{EUR/GBP} = 0.8400$

An arbitrager notices that: - Direct $\text{EUR/GBP} = 0.8400$ - Implied EUR/GBP through USD = $1.1000/1.3000 = 0.8462$

There's a profit opportunity! Buy EUR with GBP at Bank C (cheap), sell EUR for USD at Bank A, buy GBP with USD at Bank B.

Market Impact:

- Arbitrage helps ensure currency prices are consistent across markets
- Speculation provides liquidity and helps with price discovery
- Both speed up the market's response to new information
- However, excessive speculation can increase volatility

Key Distinction from Dealers:

- **Dealers** make profit from the bid-ask spread AND from taking positions
- **Speculators** make profit only from rate changes
- **Arbitragers** make profit from rate discrepancies between markets

1.7.4 Central Banks and Treasuries

Unique Participants with Policy Goals

Central banks and treasuries are fundamentally different from all other market participants because they're not motivated by profit.

Their Objectives:

1. **Acquire/Spend Foreign Exchange Reserves:** Central banks hold foreign currency reserves (typically dollars, euros, yen) to:
 - Facilitate international trade
 - Service foreign-currency debt
 - Demonstrate financial stability
2. **Influence Exchange Rates:** Practice called **foreign exchange intervention**
 - Support their own currency's value
 - Fulfill commitments under exchange rate agreements
 - Achieve domestic policy objectives

Example of Intervention:

Suppose the Mexican peso is falling rapidly:

- Market rate: MXN 22.00 = USD 1.00
- Banco de México (Mexican central bank) wants to support the peso
- They SELL U.S. dollars from their reserves
- They BUY Mexican pesos
- Increased demand for pesos can slow or reverse the decline

Willing Loss-Takers:

Here's what makes central banks unique: **They're willing to take losses on FX transactions if it serves policy goals.**

- A commercial bank seeks profit on every trade
- A speculator seeks profit on every trade
- A central bank may intentionally trade at unfavorable rates to achieve policy objectives

Example:

During a currency crisis, a central bank might:

- Pay above-market rates to attract foreign capital
- Sell forex reserves at unfavorable prices to defend the currency
- Accept substantial losses if it stabilizes the financial system

Notable Intervention Examples:

1. **Swiss National Bank (2011-2015):** Committed to preventing the Swiss franc from strengthening beyond CHF 1.20 per euro, buying vast amounts of euros (we'll see in the mini-case how this ended)
2. **Japan (Various periods):** Frequently intervenes to weaken the yen to help exporters
3. **China:** Massive interventions over decades to manage the renminbi

Modern Context:

Central bank intervention is less common than in past decades because:

- Many major currencies now float freely
- Intervention is often expensive and ineffective against market forces
- International agreements limit competitive devaluations

However, central banks remain important participants, and the threat of intervention can sometimes influence markets even without actual trading.

1.7.5 Foreign Exchange Brokers

The Intermediaries

Foreign exchange brokers are **agents** who facilitate trading between dealers without themselves becoming principals in the transaction.

How They Work:

- Maintain instant access to hundreds of dealers worldwide
- Help dealers find counterparties for trades
- Charge a small commission for this service
- Don't take positions themselves (don't buy or sell for their own account)

Why They Exist:

- **Information Efficiency:** A dealer wanting to buy €50 million could call many other banks, or call one broker who knows the market
- **Anonymity:** Sometimes dealers don't want to reveal their positions to competitors
- **Price Discovery:** Brokers help dealers find best prices quickly

Historical vs. Modern Role:

Traditional Voice Brokers: - Used telephone to connect buyers and sellers - Maintained extensive networks - Earned fees on volume

Electronic Brokers: - Platforms like EBS (Electronic Broking Services) and Reuters Dealing - Automated matching of orders - Much faster and more efficient - Lower costs - Greater transparency

Example:

A bank in Tokyo wants to sell \$100 million for yen:

Traditional: Call a voice broker → broker calls other banks → finds buyer → arranges trade → charges commission

Electronic: Input order into EBS → system automatically matches with counterparty → trade confirmed in seconds → minimal fees

Current Market:

Electronic brokers now dominate for major currency pairs. Voice brokers still handle:

- Less liquid currencies
- Very large transactions
- Complex or non-standard deals
- Emerging market currencies

1.8 Evolution of the Market

The foreign exchange market has undergone a remarkable transformation, particularly in the past 50 years. Understanding this evolution helps us appreciate both current market structure and likely future developments.

Historical Stages:

1. Fixed Exchange Rates (Bretton Woods, 1944-1971)

- Currencies pegged to U.S. dollar
- Dollar pegged to gold
- Limited currency trading—mostly for commercial purposes
- Central banks maintained fixed rates through intervention

2. Floating Rates Begin (Post-1971)

- Bretton Woods collapses
- Major currencies float
- Need for FX market explodes
- Banks become active traders
- Speculation increases dramatically

3. Telephone Era (1970s-1980s)

- All trades conducted by voice (telephone)
- Dealers called each other directly
- Brokers connected buyers and sellers by phone
- Paper-based confirmation and settlement
- High costs, slow execution

4. Electronic Revolution (1990s-2000s)

- Reuters and EBS launch electronic trading platforms
- Faster execution
- Greater transparency

- Reduced costs
- But still largely a professional/institutional market

5. Modern Era (2010-present)

- Electronic trading dominates (75%+ of volume)
- High-frequency trading and algorithms
- Retail access expands
- Mobile trading platforms
- Continuous Linked Settlement (CLS) reduces settlement risk

1.9 The Evolution of FX Trading

To truly understand how the FX market has evolved, let's examine a hypothetical trade between two dealers at three different points in time: 1985, 1990s, and 2010-today. This will illustrate the dramatic changes in technology, transparency, and market structure.

1.9.1 1985

The Scene: Two foreign exchange dealers, one in London and one in New York, working for major banks. Both are rewarded for making profits for their banks on FX trading.

The Trade:

1. **Initial Contact:** London dealer calls New York dealer on the telephone
2. **Quote Request:** "Can you give me a price on cable?" (cable = GBP/USD, from when rates were transmitted by transatlantic telegraph cable)
3. **Quote Provided:** New York dealer: "Eight-four twenty, thirty" meaning:
 - Bid: \$1.8420 per pound
 - Ask: \$1.8430 per pound
4. **Acceptance:** London dealer: "I'll sell you ten million at thirty" (selling £10 million at \$1.8430)
5. **Confirmation:** New York dealer: "Done"
6. **Recording:** Both traders write details on paper ("deal tickets")
7. **Back Office:** Paper tickets physically carried to back-office staff
8. **Documentation:** Back office creates written confirmation, sends via mail/telex
9. **Settlement:** Days later, money physically transferred between bank accounts

What They DON'T Know in 1985:

- **No real-time market data:** They don't know what rates other dealers are quoting right now

- **No recent trade history:** Unless they've called multiple banks, they don't know what rate others have just traded at
- **Limited price discovery:** Their view of the "market" is limited to the banks they've called
- **Error risk:** Handwritten tickets can be misread; numbers can be transposed
- **Settlement risk:** High risk that one party fails to deliver (Herstatt Risk—more on this soon)

Key Characteristics of 1985 Trading:

- Entirely relationship-based
- Limited transparency
- Slow execution and confirmation
- High error risk
- High settlement risk
- Closed market (only major banks participate)

1.9.2 1990s

The Scene: Same two dealers, but now technology has transformed their world.

The Trade:

1. **Computer Communication:** Dealers communicate through Reuters Dealing system (electronic chat)
2. **Market Data Available:** Both can see on their Reuters screens:
 - Recent trades in GBP/USD
 - Indicative quotes from other banks
 - Market commentary and news
3. **Quote Request:** Typed electronically: "Price for 10 cable?"
4. **Quote Provided:** "20/30" (meaning \$1.8420/\$1.8430)
 - Dealers only type the last two digits ("pips") when rate is clear
5. **Acceptance:** "Mine" or "Yours" (I'll buy or I'll sell)
6. **Automatic Recording:** System creates instant digital record
7. **Electronic Confirmation:** Generated automatically and sent electronically
8. **Back Office:** Receives trade details electronically, processes for settlement

What's Better Than 1985:

- **More market information:** Can see recent trades and quotes
- **Faster execution:** Seconds instead of minutes
- **Automatic recording:** No transcription errors
- **Electronic confirmation:** No mail delays
- **Better price discovery:** More visibility into the market

What's Still Limited:

- **Still bilateral:** Must find a willing counterparty
- **Limited access:** Smaller banks and customers can't access Reuters Dealing directly
- **Settlement risk remains:** Still settling on different days across time zones (Herstatt Risk)
- **Voice brokers still used:** For many trades, especially less liquid currencies

Electronic Brokers Emerge:

The 1990s saw the introduction of electronic brokers:

- **EBS (1992):** Launched by major banks as an electronic matching system
- **Reuters Dealing 2000-2:** Competing platform
- **Anonymous matching:** Dealers input orders, system matches automatically
- **Greater liquidity:** See more quotes simultaneously
- **Faster execution:** Trades executed in seconds

Key Characteristics of 1990s Trading:

- Hybrid electronic/voice
- Improved transparency
- Faster execution
- Lower costs
- But still largely professional market

1.9.3 2010 - Today

The Scene: The market has been revolutionized. Our two dealers are now looking at banks of computer screens, and they're competing with algorithms.

The New Reality:

1. Fragmented Trading Venues:

Dealers now access multiple platforms simultaneously:

- **Electronic Brokers:** EBS, Reuters (now Refinitiv)
- **Single-Bank Platforms:** JPMorgan's platform, Citi's platform, etc.
- **Multi-Bank Platforms:** FX Connect, Currenex, 360T
- **Prime Brokerage:** Allows hedge funds to trade directly with dealers

2. High-Frequency Trading:

- Computer algorithms execute thousands of trades per second
- Based on mathematical models
- React to market data in microseconds
- Account for significant trading volume

3. The Two-Tier Structure Has Broken Down:

Old Model (Pre-2000s):

Tier 1: Interdealer Market (banks trading with banks)

↓

Tier 2: Customer Market (banks trading with customers)

New Model (2010+):

Multiple platforms, all participants can access multiple venues:

- Retail aggregators give small customers access
- Prime brokerage gives hedge funds direct access
- Single-bank and multi-bank platforms overlap
- Electronic brokers open to more participants

4. Retail Market Revolution:

Small customers who once faced huge spreads at airport exchange kiosks can now:

- Trade online through retail platforms
- Access spreads close to institutional rates
- Use mobile apps to execute trades
- Monitor real-time market data

5. Algorithmic Trading:

Sophisticated participants use algorithms to:

- Execute large orders without moving the market
- Identify arbitrage opportunities in milliseconds
- Automatically hedge positions
- Optimize execution across multiple venues

A Modern Trade:

1. **Algorithm detects opportunity:** Computer identifies EUR/USD slightly cheaper on one platform than another
2. **Automatic execution:** Algorithm simultaneously:
 - Buys EUR on Platform A
 - Sells EUR on Platform B
 - Captures tiny profit (maybe 0.0001, but on €100 million that's \$10,000)
3. **Instant recording:** Digital record created
4. **Automatic confirmation:** Electronic confirmation sent instantly
5. **Settlement through CLS:** Continuous Linked Settlement Bank eliminates settlement risk (more on this next)

What's Dramatically Better:

- **Complete price transparency:** Real-time data on trades across all venues
- **Faster execution:** Microseconds instead of seconds

- **Lower costs:** Tight spreads even for small customers
- **Settlement risk largely eliminated:** CLS provides payment-versus-payment
- **Democratized access:** Even small participants can trade efficiently
- **Global integration:** Truly a single global market

What's More Challenging:

- **Information overload:** Too much data to process manually
- **Technology requirements:** Need sophisticated systems to compete
- **Regulatory complexity:** Multiple jurisdictions, changing rules
- **Flash crashes:** Algorithms can cause sudden, dramatic price movements
- **Cyber risk:** Hacking, system failures

Key Characteristics of Modern Trading:

- Predominantly electronic (75%+)
- Algorithm-dominated
- Fragmented across platforms
- Accessible to all participant types
- Very fast execution
- Lower costs
- Greater transparency

1.10 The Three Components of FX Trades

Today's foreign exchange trade actually involves three distinct components, each managed by different systems:

1.10.1 Component 1: The Trade Agreement

What: The actual agreement between buyer and seller on: - Currency pair - Amount
- Exchange rate - Value date (settlement date)

How: Through the platforms we just discussed: - Electronic brokers (EBS, Refinitiv) - Single-bank platforms - Multi-bank platforms - Voice brokers (for some trades)

Result: A binding agreement to exchange currencies

1.10.2 Component 2: Messaging and Payment Instructions

The System: SWIFT

What is SWIFT?

- Society for Worldwide Interbank Financial Telecommunication
- Initiated in 1973
- A secure network for financial institutions to send/receive information
- Standardized messaging format

What SWIFT Does:

- Sends payment orders
- Confirms transactions
- Provides audit trails
- Ensures secure communication

What SWIFT Does NOT Do:

- Does not transfer funds
- Does not hold funds
- Does not settle transactions
- It's messaging only!

Example SWIFT Message:

When Bank A in New York wants to pay Bank B in London:

SWIFT Message Contains:

- Sender: Bank A (NY)
- Receiver: Bank B (London)
- Amount: \$10,000,000
- Currency: USD
- Value date: October 15, 2024
- Beneficiary account details
- Purpose of payment

Importance:

- Standardized format prevents errors
- Secure encryption prevents fraud
- Permanent record for compliance
- Almost all cross-border payments use SWIFT

Security Concerns:

The 2016 Bangladesh Bank Heist (covered in your textbook's Global Finance in Practice 5.3) showed that even SWIFT can be hacked:

- Hackers installed malware on Bangladesh Central Bank's computers
- Sent fraudulent SWIFT messages to NY Fed
- Requested \$951 million transfer
- \$81 million was transferred before stopped
- Money was laundered through Philippines casinos

Lesson: Even secure systems require proper cybersecurity at endpoints.

1.10.3 Component 3: Settlement

The Problem: Herstatt Risk

Settlement risk, also known as **Herstatt Risk**, was the biggest problem in FX markets for decades.

The Bankhaus Herstatt Story (1974):

- Bankhaus Herstatt: Medium-sized German bank, active in FX trading
- June 26, 1974, 3:30 PM German time: German regulators closed the bank
- **The Problem:** Several counterparty banks had already paid Deutsche marks to Herstatt, expecting to receive U.S. dollars later that day
- **But:** Herstatt was closed before making the dollar payments
- **Result:** Herstatt's counterparties lost their Deutsche marks and received no dollars

Why Time Zones Created Risk:

- German banks close: 3:30 PM German time = 9:30 AM New York time
- Herstatt's European counterparties had made payments during European business hours
- Herstatt's New York correspondent bank (Chase Manhattan) was just opening
- Herstatt was closed before making outbound payments

Impact:

- Currency trading in New York fell more than 75% in the days following
- Banks refused to make payments until receiving confirmation of countervalue
- Created years of work on solving settlement risk

The Solution: Continuous Linked Settlement (CLS)

Introduced in 2002, CLS Bank revolutionized FX settlement:

How CLS Works:

1. **Payment versus Payment (PvP):** Both sides of the FX transaction settle simultaneously
2. **Centralized Settlement:** CLS Bank acts as intermediary
3. **Example:**
 - Bank A owes Bank B: \$10 million
 - Bank B owes Bank A: €8.4 million
 - Without CLS: Two separate payments, settlement risk
 - With CLS: Both payments made simultaneously, no settlement risk

Process:

Before CLS:

9:00 AM: Bank A pays dollars → Bank B (risk: what if Bank B doesn't pay euros?)

4:00 PM: Bank B pays euros → Bank A (or maybe not!)

With CLS:

2:00 AM: Both banks instruct CLS

2:00 AM: CLS confirms both have funds

2:01 AM: CLS simultaneously:

- Debits dollars from Bank A's account
- Credits dollars to Bank B's account
- Debits euros from Bank B's account
- Credits euros to Bank A's account

Result: Payment versus Payment - no settlement risk!

Statistics:

- As of 2015: CLS settled 18 different currencies
- Serves over 9,000 banks and institutions
- Processes over 5 million transactions per day
- Has eliminated most settlement risk in FX markets

Remaining Challenges:

- Not all currencies eligible for CLS
- Emerging market currencies often settle bilaterally
- Operational risk remains (system failures)
- Cyber risk is growing concern

1.11 FX Market Manipulation: Fixing the Fix

Following the LIBOR scandal (which we'll study in Chapter 8), similar allegations arose in foreign exchange markets in 2013-2014. This led to major investigations and reforms.

The "Fix" Problem:

What is the "Fix"?

- **London Fix:** 4:00 PM London time daily benchmark rate
- Used by institutions to value positions
- Used by index funds to set prices
- Critically important reference rate

The Manipulation:

- Traders allegedly collaborated to move rates around 4:00 PM
- Used social media, chat rooms, phone calls to coordinate
- Exchanged information about client orders
- Attempted to manipulate the fix in their favor

Evidence:

- **Steep spikes in trading** just before 4:00 PM
- Spikes not sustained afterwards
- Patterns inconsistent with random trading
- E-mail and chat evidence of collusion

Methods:

1. **Front-running client orders:** If knowing large buy order coming at 4 PM, trader buys beforehand to push price up
2. **Collusion:** Multiple traders agree to push price in same direction
3. **After-hours personal trading:** Using knowledge gained during day

Example of Trader Communications:

Trader A: "Got a big euro fix coming, you?"

Trader B: "Yeah, need euro higher, can you help?"

Trader A: "Sure, I'll buy some just before"

Consequences:

- Multiple banks fined billions of dollars
- Traders fired and some criminally prosecuted
- Greater scrutiny of FX markets
- Move toward electronic trading accelerated

The Electronic “Fix”?

Initially, increased electronic trading was seen as solution:

- Algorithms less susceptible to collusion
- More transparent pricing
- Faster execution
- Based on “reversion to mean” (stabilizing)

But...New Problems Emerged:

- Sophisticated algorithms could still manipulate
- “Spoofing”: Placing fake orders to move prices
- Flash crashes: Algorithms reacting to each other
- Detection of other traders’ mouse movements (allegedly)

Lesson: Technology alone doesn’t solve behavioral problems. Markets need:

- Strong ethical standards
- Effective surveillance
- Appropriate penalties
- Cultural change in trading rooms

1.12 The FX Global Code of Conduct 2016

Following manipulation scandals, the Bank for International Settlements (BIS) developed a global code of conduct.

Purpose:

“To promote a robust, fair, liquid, open, and appropriately transparent market in which a diverse set of Market Participants, supported by resilient infrastructure, are able to confidently and effectively transact at competitive prices that reflect available market information.”

Key Principle: This is NOT law or regulation—it’s a set of best practices that market participants voluntarily commit to follow.

The Six Leading Principles:

1. Ethics

- Behave in an ethical and professional manner
- Promote fairness and integrity
- Don’t manipulate markets
- Don’t share confidential information inappropriately

2. Governance

- Have robust policies and procedures
- Clear organizational structure
- Responsible engagement
- Proper training and supervision

3. Information Sharing

- Be clear and accurate in communications
- Protect confidential information
- Don’t front-run client orders
- Promote effective communication

4. Execution

- Exercise care when negotiating and executing transactions
- Promote fair execution
- Handle orders appropriately
- Avoid conflicts of interest

5. Risk Management and Compliance

- Promote robust control environment
- Identify, measure, monitor, manage, and report risks
- Strong compliance function
- Regular audits

6. Confirmation and Settlement

- Robust, efficient, transparent post-trade processes
- Risk-mitigating settlement
- Use of CLS where possible
- Timely confirmations

Implementation:

- Market participants publicly commit to following the Code
- Statement of Commitment published
- Annual reaffirmation required
- But no legal enforcement mechanism

Effectiveness:

Pros: - Creates industry standard - Puts pressure on firms to comply - Provides framework for compliance programs - Signals to clients that firm follows best practices

Cons: - Voluntary, no legal force - Enforcement relies on reputation - May not prevent determined bad actors - Requires constant updating

Bottom Line: The Code represents an attempt by the industry to self-regulate and restore trust after the manipulation scandals. Time will tell if it's sufficient or if more formal regulation is needed.

2 Transactions in the Foreign Exchange Market

Now that we understand the market's functions, structure, and participants, let's examine the actual transactions that occur. Foreign exchange transactions can be executed on a **spot**, **forward**, or **swap** basis. A broader definition including major derivatives would also include foreign currency **options** and **futures**.

All these transactions are defined by their **value date**—the date when the currencies are actually exchanged (settlement date). This is the organizing principle for understanding FX transactions.

2.1 Spot Transactions

Definition: A spot transaction in the interbank market is the purchase of foreign exchange with delivery and payment between banks taking place normally on the **second following business day**.

Key Characteristics:

Value Date: T+2

- Trade Date (T): Today, when agreement is made
- Value Date: Two business days later, when settlement occurs
- Exception: Canadian dollar trades with U.S. dollar settle T+1 (next business day)

Why T+2?

- Allows time for banks to process paperwork
- Confirms that both parties have funds
- Standard evolved from historical telegraph-era practices
- Now maintained as convention even though electronic systems could settle faster

Example:

Monday (T): Bank A agrees to buy €10 million from Bank B at \$1.1200/€

Tuesday (T+1): Confirmations exchanged, settlement prepared

Wednesday (T+2): Settlement:

- Bank A transfers \$11,200,000 to Bank B
- Bank B transfers €10,000,000 to Bank A

Settlement Methods:

Most dollar transactions worldwide settle through:

- **CHIPS** (Clearing House Interbank Payments System) in New York
- Calculates net balances owed between banks
- Facilitates payment by 6:00 PM in Federal Reserve Bank of New York funds
- Other central banks provide similar services for their currencies

Spot Rate Quotation:

Spot rates are quoted on an **outright basis**—all digits are stated explicitly:

EUR/USD Spot: 1.1274

Not: 74 (that would be a forward quote in "points")

Most Common Transaction:

Spot transactions are the most straightforward FX transaction:

- Simple to understand
- Immediate execution
- Quick settlement
- Used for current needs

Use Cases:

1. Immediate Commercial Needs

- Importer needs foreign currency to pay supplier now
- Exporter receives foreign currency, converts to home currency now

2. Portfolio Rebalancing

- Fund manager selling foreign stocks, converting proceeds

3. Speculation

- Trader takes position based on view of currency direction

4. Arbitrage

- Exploiting temporary price discrepancies

2.2 Forward Transactions

Definition: An **outright forward transaction** requires delivery at a future value date of a specified amount of one currency for a specified amount of another currency. The exchange rate is established at the time of agreement, but payment and delivery are not required until maturity.

Key Characteristics:

Standard Maturities:

- 1 month forward
- 2 months forward
- 3 months forward
- 6 months forward
- 12 months forward (1 year)

But: Banks will quote forwards for any maturity—even broken dates like 37 days or 5 months and 12 days.

Value Date Calculation:

Payment occurs on the **second business day after the even-month anniversary** of the trade.

Example:

Trade date: March 18

2-month forward value date: May 20 (or next business day if weekend/holiday)

Calculation:

March 18 → May 18 (two months)

May 18 → May 20 (two business days later)

Long-Dated Forwards:

According to the IMF (2014), there were **127 countries with forward markets**. While demand is heaviest for maturities of one year or less, forwards can be quoted as far as 20 years into the future for major currencies.

Forward Rates vs. Spot Rates:

The forward rate will generally differ from the spot rate. This difference reflects:

- Interest rate differential between the two currencies
- Market expectations about future spot rates
- Supply and demand for forward contracts

Example Forward Quotation:

EUR/USD Spot: 1.1200

EUR/USD 3-month forward: 1.1250

Interpretation: The forward rate is higher, meaning the euro is trading at a forward premium

How Forward Contracts Work:

Scenario: U.S. Exporter

- Today (January 1): Sign contract to sell machinery to German company
- Price: €5 million
- Payment due: July 1 (6 months)
- Current spot rate: \$1.1200/€
- 6-month forward rate: \$1.1250/€

Without Forward Contract:

- Exposed to exchange rate risk for 6 months
- If euro weakens to \$1.0500/€ by July 1:
 - Expected revenue: €5 million $\times$ \$1.1200 = \$5,600,000
 - Actual revenue: €5 million $\times$ \$1.0500 = \$5,250,000
 - Loss from FX movement: \$350,000

With Forward Contract:

- January 1: Enter forward contract to sell €5 million at \$1.1250/€ on July 1
- July 1: Regardless of spot rate, receive:
 - €5 million $\times$ \$1.1250 = \$5,625,000
- **Result:** Exchange rate risk eliminated!

Bid-Ask on Forwards:

Like spot rates, forward rates have bid-ask spreads, but often slightly wider:

3-month EUR/USD Forward:

Bid: 1.1248

Ask: 1.1252

Spread: 4 pips (0.0004)

Uses of Forward Contracts:

1. Hedging Commercial Transactions

- Exporters lock in rates for future receivables
- Importers lock in rates for future payables

2. Hedging Financial Positions

- Portfolio investors hedge foreign stock/bond holdings
- Protect against currency depreciation

3. Speculation

- Taking a view on future currency movements
- If believe euro will strengthen, buy euros forward

4. Arbitrage

- Interest rate parity ensures forward rates align with interest differentials
- Arbitrage opportunities arise if they don't

Important Note - Terminology:

“Buying euros forward for dollars” and “selling dollars forward for euros” describe the **same transaction**. Always think about which currency you're buying and which you're selling.

2.3 Swap Transactions

Definition: A swap transaction in the interbank market is the **simultaneous purchase and sale** of a given amount of foreign exchange for two different value dates. Both purchase and sale are conducted with the same counterparty.

This definition is crucial—swaps involve **TWO transactions simultaneously**.

Why Swaps Exist:

Swaps are a technique for:

- Temporarily exchanging one currency for another
- Effectively borrowing one currency while lending another
- Managing liquidity across currencies
- Adjusting the timing of FX positions

Types of Swaps:

1. Spot Against Forward (most common)
2. Forward-Forward Swaps
3. Non-deliverable Forwards (NDFs)

2.3.1 Spot Against Forward

The Most Common Swap

Structure:

- **Leg 1:** Buy currency in spot market (at spot rate)
- **Leg 2:** Simultaneously sell same amount back in forward market (at forward rate)
- Same counterparty for both legs
- No unexpected FX risk (it's a single combined transaction)

Example:

A dealer needs euros for 3 months:

Leg 1 (Spot): Buy €10 million at spot rate \$1.1200/€

- Pay: \$11,200,000
- Receive: €10,000,000
- Value date: T+2 (Wednesday)

Leg 2 (Forward): Sell €10 million at 3-month forward rate \$1.1250/€

- Receive: \$11,250,000
- Pay: €10,000,000
- Value date: 3 months + T+2

Net Result: Temporarily borrowed euros for 3 months, paid net cost of \$50,000

Economic Interpretation:

The \$50,000 cost represents the **interest rate differential** between dollars and euros for 3 months. The swap rate expresses this interest differential on a points basis rather than as an explicit interest rate.

Why Use Spot-Against-Forward Swap Instead of Just Borrowing?

- **Fully collateralized:** No credit risk beyond counterparty risk
- **Often cheaper:** Lower transaction costs than unsecured borrowing
- **Flexible:** Can be tailored to exact needs
- **Common:** Deep, liquid market

Volume:

Swap transactions and outright forwards combined make up **more than half of all foreign exchange market activity**.

2.3.2 Forward-Forward Swaps

More Sophisticated Structure

Definition: A swap where BOTH legs are forward (no spot leg).

Example:

A dealer's situation:

- Has €20 million arriving in 2 months (from a different transaction)
- Needs €20 million in 3 months (for yet another transaction)
- Wants to "bridge" this one-month gap

Forward-Forward Swap:

Leg 1: Sell €20 million for dollars, 2-month forward at \$1.1230/€

- Will receive: \$22,460,000 in 2 months

Leg 2: Buy €20 million with dollars, 3-month forward at \$1.1250/€

- Will pay: \$22,500,000 in 3 months

Net cost: \$40,000 for effectively borrowing dollars for one month

Use Cases:

- Managing timing mismatches in cash flows
- Adjusting FX position timing without closing and reopening
- Often cheaper than rolling spot positions
- Used by sophisticated corporate treasurers and banks

Relationship to Interest Rates:

The difference between the two forward rates reflects the interest rate differential between the currencies for that specific time period (1 month in our example).

Important Note:

Forward-forward swaps are why your textbook categorizes forward contracts as “foreign currency loan agreements” by banks. They effectively allow you to borrow in one currency while lending in another—exactly what a swap does.

2.3.3 Nondeliverable Forwards (NDFs)

Created for Restricted Currencies

Definition: NDFs possess the same characteristics and documentation requirements as traditional forward contracts, EXCEPT:

- Settled only in U.S. dollars
- The foreign currency is NOT delivered
- Net difference is paid in dollars

Why NDFs Exist:

Many emerging market currencies have:

- Capital controls preventing free flow across borders
- Restrictions on currency convertibility
- Limited access to forward markets for non-residents
- Governments trying to control currency values

Example Currencies Where NDFs Are Common:

- Chinese renminbi (CNY)
- Indian rupee (INR)
- Korean won (KRW)
- Brazilian real (BRL)

- Argentine peso (ARS)
- Venezuelan bolivar (VES)

How NDFs Work:

Scenario: U.S. company expects to receive 10 million Chinese renminbi in 6 months from sales in China.

Without NDF:

- Exposed to renminbi depreciation risk
- Can't access standard forward market (China has capital controls)
- Limited hedging options

With NDF:

At Inception (Today): - Enter NDF contract with bank - Notional amount: CNY 10 million - NDF Rate: 6.50 (i.e., CNY 6.50 = USD 1.00) - Maturity: 6 months - Settlement: Cash in USD only

At Maturity (6 months later):

Fixing Rate: The spot rate on the maturity date (called the “fixing rate”)

Three Possible Scenarios:

Scenario A - RMB Weakens: - Fixing rate: CNY 7.00 = USD 1.00 - Contract rate: CNY 6.50 = USD 1.00 - Calculation: - At contract rate: CNY 10 million = USD 1,538,462 - At fixing rate: CNY 10 million = USD 1,428,571 - Difference: USD 109,891 - **Settlement:** Bank pays company \$109,891 in cash - Company receives actual CNY 10 million from Chinese customer, converts at spot - Net effect: Hedged!

Scenario B - RMB Strengthens: - Fixing rate: CNY 6.00 = USD 1.00 - Contract rate: CNY 6.50 = USD 1.00 - Calculation: - At contract rate: CNY 10 million = USD 1,538,462 - At fixing rate: CNY 10 million = USD 1,666,667 - Difference: USD 128,205 - **Settlement:** Company pays bank \$128,205 in cash - Company receives actual CNY 10 million from Chinese customer, converts at spot - Net effect: Gave up some upside, but also protected against downside

Scenario C - No Change: - Fixing rate equals contract rate - No payment either direction

Key Features of NDFs:

1. **No Physical Delivery:** Foreign currency never changes hands in the NDF contract itself
2. **Dollar Settlement Only:** Net difference paid in USD
3. **Based on Fixing Rate:** Uses official or representative spot rate on maturity date
4. **Offshore Trading:** Traded outside the country of the currency (e.g., Chinese renminbi NDFs traded in Hong Kong, Singapore, London, New York—NOT in mainland China)
5. **Beyond Government Control:** This is precisely why they exist—to allow hedging and speculation in currencies with capital controls

NDF Pricing:

How do banks price NDFs when there's no accessible money market for interest rates?

- **If money markets exist:** Use interest rate parity (covered in Chapter 6)
- **If money markets don't exist:** More speculative
 - Based on what dealers think future spot rate will be
 - Supply and demand in NDF market itself
 - Fundamental analysis of country economy
 - Political risk assessment

Problems with NDFs:

The Fixing Problem:

In times of crisis, governments may:

- Suspend spot trading (so no “fixing rate” available)
- Create multiple exchange rates (which one to use?)
- Impose additional capital controls

Real Example - Venezuela 2003:

- Government suspended FX trading during crisis
- No official spot rate available
- NDFs couldn't be settled (no fixing rate!)
- Eventually government announced new rate, but still no actual trading
- Created legal disputes between banks and clients

Real Example - Argentina 2001:

- Argentina abandoned 1:1 peso-dollar peg
- Peso fell dramatically
- NDF market had predicted this (trading at weaker rates)
- Government was angry that NDFs undermined their currency defense

Usage Statistics:

- Created in early 1990s
- Initially focused on Latin American currencies
- Asian NDFs now dominate (especially Chinese renminbi)
- Estimated 70%+ of NDF trading is for speculation (not hedging)

Important Context:

NDFs have proven to be an imperfect substitute for traditional forwards, but they're often the only hedging tool available for restricted currencies. They create a “gray market” for currencies that governments are trying to control—sometimes helping price discovery, sometimes undermining government policies.

2.4 Size of the Foreign Exchange Market

Let's put some numbers on this massive market.

Current Size (2019 BIS Survey):

- **Daily trading volume: \$6.6 trillion per day**
- Up 30% from 2016
- Up over 1,000% since 1989

For perspective:

- Global stock markets: roughly \$100-150 billion daily trading
- The FX market is about 40-50 times larger!
- Annual global GDP: about \$90 trillion
- FX market trades equivalent of entire global GDP every 2 weeks

Transaction Breakdown (2019):

- **Spot transactions:** \$2.0 trillion per day (30%)
- **Outright forwards:** \$1.0 trillion per day (15%)
- **FX swaps:** \$3.2 trillion per day (49%)
- **Options and other derivatives:** \$0.4 trillion per day (6%)

Growth Trends:

Looking at the chart in your textbook:

- **1989:** \$590 billion per day
- **2001:** \$1,239 billion per day (crisis caused brief decline)
- **2007:** \$3,324 billion per day (pre-financial crisis peak)
- **2010:** \$3,973 billion per day (recovery from crisis)
- **2013:** \$5,357 billion per day
- **2016:** \$5,066 billion per day (slight decline—only happened twice)
- **2019:** \$6,590 billion per day (strong recovery)

Interesting Observations:

1. **2001 Decline:** Global recession temporarily reduced trading
2. **2008-2009 Crisis:** Market actually grew! Unlike many markets, crisis increased FX volatility and trading
3. **2016 Decline:** First significant decline since 2001
 - Spot trading fell for first time
 - Possibly due to lower volatility
 - Consolidated among fewer, larger dealers
4. **2019 Recovery:** Strong 30% increase
 - Electronic trading growth
 - More profit-seekers entering market

- Big data and algorithmic trading expanding

What Drives Growth?

- **Globalization:** More international trade and investment
- **Technology:** Electronic trading reduces costs
- **Speculation:** More participants seeking profits
- **Derivatives:** Growing use of FX derivatives
- **Emerging Markets:** New currencies becoming tradable
- **Algorithmic Trading:** High-frequency trading increasing volume

2.5 Geographical Distribution

Where does all this trading occur? While the market is global and electronic, certain cities dominate.

Top Trading Centers (2019):

1. **United Kingdom (London):** 43% of global trading
 - \$3.6 trillion per day
 - Dominant for centuries
 - Brexit concerns haven't diminished its role (yet)
2. **United States (New York):** 17% of global trading
 - \$1.4 trillion per day
 - Second largest
 - Overlap with London creates peak liquidity
3. **Singapore:** 8% of global trading
 - \$640 billion per day
 - Growing Asian hub
 - Gateway to Asian markets
4. **Hong Kong:** 7.6% of global trading
 - \$632 billion per day
 - China's international financial center
 - Renminbi trading hub
5. **Japan (Tokyo):** 4.5% of global trading
 - \$376 billion per day
 - Declining share (was 10%+ in 1990s)
 - Yen still major currency

Top 10 Combined: These 10 centers account for 91% of all FX trading globally.

Regional Trends:

Europe: Still dominant, but: - London maintains position despite Brexit - Euro introduction reduced intra-European FX needs - Frankfurt, Paris, Zurich all significant

Asia: Rapidly growing: - Singapore challenging Tokyo - Hong Kong crucial for China access - Shanghai growing with renminbi internationalization - Sydney active during Asian session

Americas: Relatively stable: - New York firmly second - Chicago important for futures - Toronto, São Paulo regional centers

The London-New York Axis:

Together, these two centers account for **over 60% of global FX trading**. The 5-hour overlap in their business days (8 AM - 12 PM New York time = 1 PM - 5 PM London time) is when:

- Most liquidity available
- Tightest spreads
- Highest volume
- Most active price discovery

2.6 Currency Composition

Finally, which currencies are actually traded?

Top Currency Pairs (2019):

Remember: Each trade involves TWO currencies, so percentages sum to 200%.

1. **EUR/USD:** 24.0% of all trades
 - The most liquid currency pair
 - Tightest spreads
 - Used as reference for many other pairs
2. **USD/JPY:** 13.2% of all trades
 - Second most liquid
 - Important Asian trading
 - “Safe haven” status
3. **GBP/USD:** 9.6% of all trades
 - “Cable” (from telegraph days)
 - Brexit has increased volatility
4. **AUD/USD:** 5.4% of all trades
 - Growing importance
 - Commodity currency
5. **USD/CNY:** 4.1% of all trades
 - Rapid growth
 - Was barely trading in 2001
 - Will likely continue growing

Dollar Dominance:

The U.S. dollar is involved in **88% of all FX transactions**. Even when trading euros for yen, traders often:

1. Sell euros for dollars
2. Buy yen with dollars

Rather than direct EUR/JPY trade. Why?

- Most liquid dollar markets
- Easier to find counterparties
- Historical convention
- Dollar is global reserve currency

Euro:

- Involved in 32% of transactions
- Second most important currency
- Eliminated need for intra-European currency trading

Emerging Market Currencies - Rising Stars:

- **Chinese Renminbi:** 4.1% (up from 0.8% in 2010)
- **Mexican Peso:** 1.6%
- **Korean Won:** 1.9%
- **Indian Rupee:** 1.7%
- **Brazilian Real:** 1.0%
- **Russian Ruble:** 1.0%

These currencies have grown from negligible to significant over the past two decades.

A Note on Market Concentration:

The top 10 currency pairs account for roughly 60% of all FX trading. This concentration means:

- Very liquid, tight spreads for major pairs
- Much wider spreads for exotic pairs
- Some currencies trade very infrequently
- “Exotic” currency pairs may require special attention

3 Foreign Exchange Rates and Quotations

Now we come to one of the most practical and potentially confusing topics in international finance: **How are exchange rates actually quoted?** Understanding quotation conventions is absolutely essential for:

- Reading financial news correctly
- Calculating currency conversions
- Determining profits/losses
- Identifying arbitrage opportunities
- Understanding forward premiums and discounts

Fundamental Definitions:

- **Foreign Exchange Rate:** The price of one currency expressed in terms of another currency
- **Foreign Exchange Quotation:** A statement of willingness to buy or sell at an announced rate

Think of pricing an orange:

If the price of an orange is \$1.20: - The “price” is \$1.20 - The “unit” is the orange

Same concept with currencies, but we’re pricing one currency in terms of another currency.

3.1 Currency Symbols

Before we dive into quotations, let’s clarify currency notation systems.

Two Systems in Use:

1. **Traditional Currency Symbols** (like \$, €, £)
2. **ISO 4217 Codes** (like USD, EUR, GBP)

Major Currencies:

Currency	Traditional Symbol	ISO 4217 Code
U.S. dollar	\$	USD
European euro	€	EUR
British pound	£	GBP
Japanese yen	¥	JPY
Mexican peso	Ps	MXN
Swiss franc	Fr. or CHF	CHF
Canadian dollar	C\$	CAD
Australian dollar	A\$	AUD

Modern Practice:

- **Electronic trading:** Always uses ISO codes (EUR/USD, USD/JPY)
- **Financial press:** Mixed—European/American publications often use traditional symbols, Asian/Middle Eastern often use ISO codes
- **Business communications:** Increasingly standardized on ISO codes
- **Retail/consumer:** Still uses traditional symbols (currency notes, retail exchange)

Complete List: Your textbook has a comprehensive list of currency symbols and ISO codes at the end. I recommend familiarizing yourself with the major ones.

This Course: We’ll primarily use ISO codes (USD, EUR, GBP, JPY, etc.) because that’s what professionals use, but you should recognize both systems.

3.2 Exchange Rate Quotes

Now for the mechanics of currency quotations. **Pay close attention here—this is where many students get confused initially, but once you understand the logic, it becomes straightforward.**

The Basic Structure:

Every currency quotation involves TWO currencies:

CUR1/CUR2

Key Terminology:

- **Base Currency** (or **Unit Currency**): The currency on the LEFT of the slash
 - This currency is always expressed as ONE unit
 - This is the “product” being priced
- **Price Currency** (or **Quote Currency**): The currency on the RIGHT of the slash
 - This is the “price”
 - This tells you how much of the price currency equals one unit of the base currency

Example:

EUR/USD 1.2174

Reading this:

- **EUR** is the base currency (the “product”)
- **USD** is the price currency (the “price”)
- One euro equals 1.2174 U.S. dollars
- Or: EUR 1.00 = USD 1.2174

Critical Rule to Remember:

The currency on the left of the slash is ALWAYS the base currency and is ALWAYS a single unit.

If you remember this rule, you can avoid most confusion.

Practice Reading:

Let's practice with several examples:

```
# Example 1: EUR/USD 1.2174
cat("EUR/USD 1.2174 means:\n")
```

EUR/USD 1.2174 means:

```
cat("  EUR 1.00 = USD 1.2174\n")
```

EUR 1.00 = USD 1.2174

```
cat("  One euro equals 1.2174 dollars\n\n")
```

One euro equals 1.2174 dollars

```
# Example 2: USD/JPY 118.27
```

```
cat("USD/JPY 118.27 means:\n")
```

USD/JPY 118.27 means:

```
cat("  USD 1.00 = JPY 118.27\n")
```

USD 1.00 = JPY 118.27

```
cat("  One dollar equals 118.27 yen\n\n")
```

One dollar equals 118.27 yen

```
# Example 3: GBP/USD 1.5552
```

```
cat("GBP/USD 1.5552 means:\n")
```

GBP/USD 1.5552 means:

```
cat("  GBP 1.00 = USD 1.5552\n")
```

GBP 1.00 = USD 1.5552

```
cat("  One pound equals 1.5552 dollars\n")
```

One pound equals 1.5552 dollars

Converting Values:

If you want to know how many dollars you need to buy 5,000 euros at EUR/USD 1.2174:

```
euros <- 5000
```

```
rate <- 1.2174 # USD per EUR
```

```
dollars_needed <- euros * rate
```

```
cat("To buy EUR", euros, "at rate", rate, ":\n")
```


To buy EUR 5000 at rate 1.2174 :

```
cat("You need USD", dollars_needed, "\n")
```

You need USD 6087

3.3 Market Conventions

The international currency market has evolved over centuries, and certain conventions have become standard. Understanding these conventions is essential for reading financial news and trading currencies.

3.3.1 European Terms

Definition: European terms means quoting the amount of a specific currency per one U.S. dollar.

In other words: The USD is the base currency, and the foreign currency is the price currency.

Format: USD/Foreign Currency

Examples:

- USD/JPY 118.27 (118.27 yen per dollar)
- USD/MXN 19.94 (19.94 pesos per dollar)
- USD/CHF 0.9819 (0.9819 Swiss francs per dollar)

Historical Context:

For most of the past 60+ years, the U.S. dollar has been the base currency for most quotations globally. This reflected:

- Dollar's status as global reserve currency
- Bretton Woods legacy (currencies pegged to dollar)
- London and New York dominance of FX markets
- Convenience of expressing all currencies against one standard

Example of How This Works:

A trader in Zurich (Switzerland) wants a quote on Norwegian krone. The Norwegian trader would quote:

USD/NOK 8.2727

NOT CHF/NOK (Swiss francs per Norwegian krone)

The convention is to quote everything against the dollar as the standard reference point.

3.3.2 American Terms

Definition: American terms means quoting the U.S. dollar price of one unit of foreign currency.

In other words: The foreign currency is the base currency, and the USD is the price currency.

Format: Foreign Currency/USD

Examples:

- EUR/USD 1.2174 (1.2174 dollars per euro)
- GBP/USD 1.5552 (1.5552 dollars per pound)
- AUD/USD 0.7829 (0.7829 dollars per Australian dollar)

The Major Exceptions:

There are significant exceptions to the European terms convention:

1. **Euro (EUR):** Always quoted EUR/USD (American terms)
2. **British Pound (GBP):** Always quoted GBP/USD (American terms)
3. **Australian Dollar (AUD):** Typically quoted AUD/USD (American terms)
4. **New Zealand Dollar (NZD):** Typically quoted NZD/USD (American terms)

Why the Exceptions?

British Pound:

Historical tradition! For centuries:

- Britain was the world's financial center (London)
- The pound was the reserve currency
- Foreign exchange prices were stated in foreign currency per pound
- The pound consisted of 20 shillings of 12 pence each—difficult for calculations
- Convention was to quote foreign currency per pound rather than fractions of pounds
- Even after decimalization in 1971, convention remained

Euro:

Practical transition:

- Euro replaced domestic currencies (Deutsche mark, French franc, etc.)
- Those currencies were quoted as domestic currency per euro (e.g., DM 1.95 = EUR 1.00)
- To make transition easy, continued quoting domestic currency per euro
- For USD: Quoted as USD per EUR
- Made sense to continue American terms

Australian and New Zealand Dollars:

- Followed British Commonwealth conventions
- Both were once part of sterling area

Also American Terms:

- Foreign currency **options** (we'll study in Chapter 7)

- Foreign currency **futures** (also Chapter 7)
- **Retail markets** dealing with tourists and personal remittances

3.3.3 Currency Nicknames

Professional traders use nicknames for efficiency and because it's part of trading culture.

Common Nicknames:

Currency Pair	Nickname	Origin
GBP/USD	"Cable"	From transatlantic telegraph cable days
USD/CAD	"Loonie"	Canadian \$1 coin has a loon (water bird)
NZD	"Kiwi"	New Zealand national bird
AUD	"Aussie"	Short for Australian
CHF	"Swissie"	Swiss franc
SGD	"Sing dollar"	Singapore dollar

"Cable" History:

The GBP/USD pair is called "cable" because in the 1800s, exchange rates between London and New York were transmitted via the transatlantic telegraph cable. The name stuck!

Professional Communication:

Trader 1: "What's cable trading at?"

Trader 2: "Fifty-five, sixty" (meaning 1.5550/1.5560)

You'll encounter these nicknames in financial news and trading contexts.

3.3.4 Direct and Indirect Quotations

Now we introduce another set of terms that can cause confusion: **direct** and **indirect**. These terms are **perspective-dependent**—they change based on whose viewpoint you're taking.

Critical Difference from American/European Terms:

- **American/European terms:** Based on whether USD is price or base currency (objective definition)
- **Direct/Indirect:** Based on YOUR home currency (subjective definition)

Definitions:

Direct Quote: The price of a foreign currency in **domestic currency units** (home currency is the price currency)

Indirect Quote: The price of the **domestic currency** in foreign currency units (home currency is the base currency)

Example - U.S. Perspective:

EUR/USD 1.2174

For a U.S. resident:

- Home currency = USD
- Foreign currency = EUR
- This is a **DIRECT** quote on the euro (home currency price of one unit of foreign)
- This is an **INDIRECT** quote on the dollar (foreign currency price of one unit of home)

Example - French Perspective:

EUR/USD 1.2174

For a French resident:

- Home currency = EUR
- Foreign currency = USD
- This is an **INDIRECT** quote on the euro (foreign currency price of one unit of home)
- This is a **DIRECT** quote on the dollar (home currency price of one unit of foreign)

Same Quote, Different Interpretation!

This is why direct/indirect can be confusing—the same number means different things to different people based on their home currency.

Classic Example from Textbook:

Woman in Paris:

EUR 0.8214 = USD 1.00

For her: - This is a **direct quote** on the dollar (home currency euros per foreign currency dollar) - “It will cost me 0.8214 euros to get one dollar”

Man in New York:

USD 1.2174 = EUR 1.00

For him: - This is a **direct quote** on the euro (home currency dollars per foreign currency euro) - “I will pay 1.2174 dollars per euro”

Note: These two quotes are reciprocals of each other:

```
paris_quote <- 0.8214 # EUR per USD
new_york_quote <- 1.2174 # USD per EUR

# Check that they're reciprocals
cat("Paris quote (EUR per USD):", paris_quote, "\n")
```

Paris quote (EUR per USD): 0.8214

```
cat("New York quote (USD per EUR):", new_york_quote, "\n")
```

New York quote (USD per EUR): 1.2174

```
cat("1 / Paris quote:", 1/paris_quote, "\n")
```

1 / Paris quote: 1.217434

```
cat("1 / New York quote:", 1/new_york_quote, "\n")
```

1 / New York quote: 0.8214227

```
cat("\nThey're reciprocals (within rounding)!\n")
```

They're reciprocals (within rounding)!

Bottom Line:

- When reading financial news, always identify what currencies are involved
- Figure out which is base and which is quote
- If considering direct/indirect, always specify: “from whose perspective?”

This Course:

We’ll primarily use the objective terms (base currency/quote currency, American terms/European terms) because they don’t depend on perspective.

3.3.5 Bid and Ask Rates

In the real market, you never see just ONE exchange rate. You always see **two rates**:

- A rate at which dealers will BUY (bid)
- A rate at which dealers will SELL (ask or offer)

Key Definitions:

Bid: The price (exchange rate) in one currency at which a dealer will **BUY** another currency

Ask (or Offer): The price (exchange rate) at which a dealer will **SELL** the other currency

The Spread: The difference between bid and ask—this is the dealer’s profit

Example:

EUR/USD 1.2170/1.2178

or written as:

EUR/USD 1.2170/78 (last two digits only for ask)

or written as:

EUR/USD 1.2174 ± 4 pips

Interpreting This:

- **Bid: 1.2170** → Dealer will BUY euros at this rate (paying \$1.2170 per euro)
- **Ask: 1.2178** → Dealer will SELL euros at this rate (receiving \$1.2178 per euro)
- **Spread: 0.0008 or 8 pips** → Dealer's profit margin

From YOUR Perspective (Customer):

- **You can SELL 1 euro FOR \$1.2170** (you're selling at the dealer's bid)
- **You can BUY 1 euro FOR \$1.2178** (you're buying at the dealer's ask)

The dealer always wins: They buy low (bid) and sell high (ask).

Example Transaction:

```
# You want to buy EUR 1,000,000
# Quote: EUR/USD 1.2170/1.2178

euros_to_buy <- 1000000
ask_rate <- 1.2178 # You buy at dealer's ask price

cost_in_dollars <- euros_to_buy * ask_rate

cat("To buy EUR", euros_to_buy, "\n")
```

To buy EUR 1e+06

```
cat("At ask rate:", ask_rate, "\n")
```

At ask rate: 1.2178

```
cat("You pay: USD", format(cost_in_dollars, big.mark=",", scientific=FALSE), "\n\n")
```

You pay: USD 1,217,800

```
# Now imagine selling those euros back immediately
bid_rate <- 1.2170 # You sell at dealer's bid price

dollars_received <- euros_to_buy * bid_rate

cat("If you immediately sell EUR", euros_to_buy, "\n")
```

If you immediately sell EUR 1e+06

```
cat("At bid rate:", bid_rate, "\n")
```

At bid rate: 1.217

```
cat("You receive: USD", format(dollars_received, big.mark=",", scientific=FALSE), "\n\n")
```

You receive: USD 1,217,000

```
# Your loss
loss <- cost_in_dollars - dollars_received

cat("Your loss (dealer's profit):", format(loss, big.mark=",", scientific=FALSE), "\n")
```

Your loss (dealer's profit): 800

```
cat("This is the spread:", loss / euros_to_buy, "per euro\n")
```

This is the spread: 8e-04 per euro

Bid-Ask Spread Determinants:

Spreads vary based on:

1. **Liquidity:** Major pairs (EUR/USD) have tiny spreads (2-4 pips). Exotic pairs have much wider spreads.
2. **Volatility:** Higher volatility = wider spreads (more risk for dealer)
3. **Volume:** Higher volume = narrower spreads (economy of scale)
4. **Time of Day:** Spreads widen during off-hours (less liquidity)
5. **Market Conditions:** During crises, spreads can explode

Typical Spreads (Interbank Market):

- EUR/USD: 1-4 pips (0.0001-0.0004)

- USD/JPY: 1-4 pips
- GBP/USD: 2-5 pips
- Exotic pairs: 50-200+ pips

Retail Spreads: Much wider! Airport currency exchange might have spreads of 500-1000 pips (5-10%).

Important Note on Financial News:

When The Wall Street Journal or Bloomberg reports exchange rates, they typically report either:

- The bid rate only
- The mid-point (average of bid and ask)
- Without specifying which!

Example from textbook:

The Wall Street Journal
EUR/USD: 1.2170

This is probably the bid or mid-rate, but not specified.

3.4 Cross Rates

Definition: A **cross rate** is an exchange rate between two currencies, calculated from their common relationships with a third currency.

Why Cross Rates Exist:

Many currency pairs are **only inactively traded**. Rather than maintaining a direct market, their exchange rate is determined through a widely-traded third currency (usually the USD).

Example:

Suppose a Mexican importer needs Japanese yen to pay for purchases in Tokyo.

Problem: The MXN/JPY direct market is very thin (not many people trade Mexican pesos for Japanese yen directly).

Solution: Use cross rate through USD:

Step 1: Look up commonly quoted rates:

USD/JPY 108.84 (yen per dollar)
USD/MXN 19.9392 (pesos per dollar)

Step 2: Calculate cross rate:


```
# Given rates
usd_jpy <- 108.84 # JPY per USD
usd_mxn <- 19.9392 # MXN per USD

# Calculate JPY per MXN (how many yen per one peso)
jpy_per_mxn <- usd_jpy / usd_mxn

cat("Given:\n")
```

Given:

```
cat("  USD/JPY:", usd_jpy, "(yen per dollar)\n")
```

USD/JPY: 108.84 (yen per dollar)

```
cat("  USD/MXN:", usd_mxn, "(pesos per dollar)\n\n")
```

USD/MXN: 19.9392 (pesos per dollar)

```
cat("Cross rate calculation:\n")
```

Cross rate calculation:

```
cat("  JPY per MXN = (JPY per USD) / (MXN per USD)\n")
```

JPY per MXN = (JPY per USD) / (MXN per USD)

```
cat("  JPY per MXN =", usd_jpy, "/", usd_mxn, "\n")
```

JPY per MXN = 108.84 / 19.9392

```
cat("  JPY per MXN =", round(jpy_per_mxn, 4), "\n\n")
```

JPY per MXN = 5.4586

```
cat("Interpretation:\n")
```

Interpretation:

```
cat("  MXN 1.00 = JPY", round(jpy_per_mxn, 4), "\n")
```

MXN 1.00 = JPY 5.4586

```
cat("  One Mexican peso equals", round(jpy_per_mxn, 4), "Japanese yen\n")
```

One Mexican peso equals 5.4586 Japanese yen

Alternative Calculation (Reciprocal):

```
# Calculate MXN per JPY (how many pesos per one yen)
mxn_per_jpy <- usd_mxn / usd_jpy

cat("Alternative calculation:\n")
```

Alternative calculation:

```
cat("  MXN per JPY = (MXN per USD) / (JPY per USD)\n")
```

MXN per JPY = (MXN per USD) / (JPY per USD)

```
cat("  MXN per JPY =", usd_mxn, "/", usd_jpy, "\n")
```

MXN per JPY = 19.9392 / 108.84

```
cat("  MXN per JPY =", round(mxn_per_jpy, 6), "\n\n")
```

MXN per JPY = 0.183197

```
cat("Interpretation:\n")
```

Interpretation:

```
cat("  JPY 1.00 = MXN", round(mxn_per_jpy, 6), "\n")
```

JPY 1.00 = MXN 0.183197

```
cat("  One Japanese yen equals", round(mxn_per_jpy, 6), "Mexican pesos\n")
```

One Japanese yen equals 0.183197 Mexican pesos

More Complex Example:

EUR/GBP cross rate from EUR/USD and GBP/USD:

```
# Given rates
eur_usd <- 1.2174 # USD per EUR
gbp_usd <- 1.5552 # USD per GBP

# Calculate EUR per GBP
eur_per_gbp <- gbp_usd / eur_usd

cat("Given:\n")
```

Given:

```
cat(" EUR/USD:", eur_usd, "(dollars per euro)\n")
```

EUR/USD: 1.2174 (dollars per euro)

```
cat(" GBP/USD:", gbp_usd, "(dollars per pound)\n\n")
```

GBP/USD: 1.5552 (dollars per pound)

```
cat("Cross rate calculation:\n")
```

Cross rate calculation:

```
cat(" EUR per GBP = (USD per GBP) / (USD per EUR)\n")
```

EUR per GBP = (USD per GBP) / (USD per EUR)

```
cat(" EUR per GBP =", gbp_usd, "/", eur_usd, "\n")
```

EUR per GBP = 1.5552 / 1.2174

```
cat(" EUR per GBP =", round(eur_per_gbp, 4), "\n\n")
```

EUR per GBP = 1.2775

```
cat("Interpretation:\n")
```

Interpretation:

```
cat("  GBP 1.00 = EUR", round(eur_per_gbp, 4), "\n")
```

GBP 1.00 = EUR 1.2775

```
cat("  One British pound equals", round(eur_per_gbp, 4), "euros\n")
```

One British pound equals 1.2775 euros

Cross Rate Matrices:

Financial newspapers often publish cross-rate tables showing how major currencies relate to each other. Your textbook has an example from Bloomberg.

3.5 Intermarket Arbitrage

When cross rates calculated from actual quotes **don't match** direct quotes between currencies, an arbitrage opportunity exists. This is called **triangular arbitrage**.

The Concept:

If exchange rates across different markets aren't consistent, you can:

1. Start with one currency
2. Exchange it for a second currency
3. Exchange that for a third currency
4. Exchange back to original currency
5. End up with MORE than you started with!

This is “risk-free” profit (assuming simultaneous execution).

Classic Example from Textbook:

Given Quotes:

Citibank (NYC): USD 1.3297 = EUR 1.00

Barclays (London): USD 1.5585 = GBP 1.00

Dresdner Bank (Frankfurt): EUR 1.1722 = GBP 1.00

Calculate Implied Cross Rate:

From the first two quotes, the implied EUR/GBP rate should be:

```
# Given rates
usd_per_eur <- 1.3297 # Citibank
usd_per_gbp <- 1.5585 # Barclays
eur_per_gbp_quoted <- 1.1722 # Dresdner actual quote

# Calculate implied EUR per GBP
eur_per_gbp_implied <- usd_per_gbp / usd_per_eur

cat("Given quotes:\n")
```

Given quotes:

```
cat(" Citibank: USD", usd_per_eur, "= EUR 1.00\n")
```

Citibank: USD 1.3297 = EUR 1.00

```
cat(" Barclays: USD", usd_per_gbp, "= GBP 1.00\n")
```

Barclays: USD 1.5585 = GBP 1.00

```
cat(" Dresdner: EUR", eur_per_gbp_quoted, "= GBP 1.00\n\n")
```

Dresdner: EUR 1.1722 = GBP 1.00

```
cat("Implied EUR/GBP from first two quotes:\n")
```

Implied EUR/GBP from first two quotes:

```
cat(" EUR per GBP = USD per GBP / USD per EUR\n")
```

EUR per GBP = USD per GBP / USD per EUR

```
cat(" EUR per GBP =", usd_per_gbp, "/", usd_per_eur, "\n")
```

EUR per GBP = 1.5585 / 1.3297

```
cat(" EUR per GBP =", round(eur_per_gbp_implied, 4), "\n\n")
```

EUR per GBP = 1.1721

```
cat("Comparison:\n")
```

Comparison:

```
cat("  Implied rate: EUR", round(eur_per_gbp_implied, 4), "= GBP 1.00\n")
```

Implied rate: EUR 1.1721 = GBP 1.00

```
cat("  Dresdner rate: EUR", eur_per_gbp_quoted, "= GBP 1.00\n")
```

Dresdner rate: EUR 1.1722 = GBP 1.00

```
cat("  Difference:", round(eur_per_gbp_implied - eur_per_gbp_quoted, 4), "\n\n")
```

Difference: -1e-04

```
cat("There is a discrepancy! Arbitrage opportunity exists.\n")
```

There is a discrepancy! Arbitrage opportunity exists.

The Arbitrage:

```
# Start with $1,000,000
```

```
starting_usd <- 1000000
```

```
cat("TRIANGULAR ARBITRAGE EXECUTION\n")
```

TRIANGULAR ARBITRAGE EXECUTION

```
cat("=====\n\n")
```

=====

```
cat("Step 1: Start with USD", format(starting_usd, big.mark=","), "\n\n")
```

Step 1: Start with USD 1e+06

```
# Step 2: Sell USD for GBP at Barclays
gbp_received <- starting_usd / usd_per_gbp

cat("Step 2: Sell USD for GBP at Barclays\n")
```

Step 2: Sell USD for GBP at Barclays

```
cat("  Rate: USD", usd_per_gbp, "= GBP 1.00\n")
```

Rate: USD 1.5585 = GBP 1.00

```
cat("  GBP received:", format(round(gbp_received, 2), big.mark=","), "\n\n")
```

GBP received: 641,642.6

```
# Step 3: Sell GBP for EUR at Dresdner
eur_received <- gbp_received * eur_per_gbp_quoted

cat("Step 3: Sell GBP for EUR at Dresdner\n")
```

Step 3: Sell GBP for EUR at Dresdner

```
cat("  Rate: EUR", eur_per_gbp_quoted, "= GBP 1.00\n")
```

Rate: EUR 1.1722 = GBP 1.00

```
cat("  EUR received:", format(round(eur_received, 2), big.mark=","), "\n\n")
```

EUR received: 752,133.5

```
# Step 4: Sell EUR for USD at Citibank
final_usd <- eur_received * usd_per_eur

cat("Step 4: Sell EUR for USD at Citibank\n")
```

Step 4: Sell EUR for USD at Citibank

```
cat("  Rate: USD", usd_per_eur, "= EUR 1.00\n")
```

Rate: USD 1.3297 = EUR 1.00

```
cat("  USD received:", format(round(final_usd, 2), big.mark=","), "\n\n")
```

USD received: 1,000,112

```
# Calculate profit
profit <- final_usd - starting_usd
```

```
cat("=====\n")
```

=====

```
cat("RESULT:\n")
```

RESULT:

```
cat("  Started with: USD", format(starting_usd, big.mark=","), "\n")
```

Started with: USD 1e+06

```
cat("  Ended with: USD", format(round(final_usd, 2), big.mark=","), "\n")
```

Ended with: USD 1,000,112

```
cat("  Profit: USD", format(round(profit, 2), big.mark=","), "\n")
```

Profit: USD 111.86

```
cat("  Return:", round((profit/starting_usd)*100, 4), "%\n")
```

Return: 0.0112 %

Important Points:

1. **Risk-Free (Theoretically):** If executed simultaneously, no currency risk
2. **Requires Speed:** Arbitrage opportunities disappear quickly as others exploit them
3. **Transaction Costs:** Real-world bid-ask spreads might eliminate small arbitrage profits
4. **Competitive Market:** The fact that these opportunities are rare and quickly eliminated demonstrates that FX markets are very efficient
5. **Market Correction:** Arbitrage activity itself corrects the mispricing:
 - Increased demand for underpriced currency pushes its price up
 - Increased supply of overpriced currency pushes its price down
 - Process continues until rates are back in alignment

3.6 Forward Quotations

Finally, let's examine how forward exchange rates are quoted.

Key Difference from Spot:

Spot rates are quoted **outright** (all digits stated explicitly):

EUR/USD Spot: 1.1274

Forward rates are typically quoted in **points** (or **pips**):

EUR/USD 3-month forward: 54 points

What are "Points"?

Points (or **pips**) represent the **difference** between the forward rate and the spot rate, expressed in the last digits of the quotation.

Example:

EUR/USD Spot: 1.0897 / 1.0901 (bid / ask)

EUR/USD 3-month forward: 53 / 54 (points)

Calculate Outright Forward:

```
# Spot rates
spot_bid <- 1.0897
spot_ask <- 1.0901

# Forward points (divide by 10,000 for EUR/USD)
forward_points_bid <- 53
forward_points_ask <- 54

# Calculate outright forward rates
forward_bid <- spot_bid + (forward_points_bid / 10000)
forward_ask <- spot_ask + (forward_points_ask / 10000)

cat("Given:\n")
```

Given:

```
cat(" EUR/USD Spot Bid:", spot_bid, "\n")
```

EUR/USD Spot Bid: 1.0897

```
cat("  EUR/USD Spot Ask:", spot_ask, "\n")
```

EUR/USD Spot Ask: 1.0901

```
cat("  3-month Forward Points:", forward_points_bid, "/", forward_points_ask, "\n\n")
```

3-month Forward Points: 53 / 54

```
cat("Outright Forward Calculation:\n")
```

Outright Forward Calculation:

```
cat("  Forward Bid = Spot Bid + (Points / 10000)\n")
```

Forward Bid = Spot Bid + (Points / 10000)

```
cat("  Forward Bid =", spot_bid, "+", forward_points_bid, "/ 10000\n")
```

Forward Bid = 1.0897 + 53 / 10000

```
cat("  Forward Bid =", forward_bid, "\n\n")
```

Forward Bid = 1.095

```
cat("  Forward Ask = Spot Ask + (Points / 10000)\n")
```

Forward Ask = Spot Ask + (Points / 10000)

```
cat("  Forward Ask =", spot_ask, "+", forward_points_ask, "/ 10000\n")
```

Forward Ask = 1.0901 + 54 / 10000

```
cat("  Forward Ask =", forward_ask, "\n\n")
```

Forward Ask = 1.0955

```
cat("Result:\n")
```

Result:

```
cat(" EUR/USD 3-month Forward:", forward_bid, "/", forward_ask, "\n")
```

EUR/USD 3-month Forward: 1.095 / 1.0955

Why Use Points?

- **Efficiency:** Traders don't need to state full numbers repeatedly
- **Focus on Change:** Points highlight how forward differs from spot
- **Convention:** It's simply how the market operates

Bid and Ask on Forward Points:

Notice the forward quotation has two numbers: **53/54**

- **First number (53):** Points away from **spot bid**
- **Second number (54):** Points away from **spot ask**

Reading Forward Quotes:

If forward points are POSITIVE (as above): - Add points to spot rate - Forward rate is higher than spot - Foreign currency trades at a **forward premium**

If forward points are NEGATIVE: - Subtract points from spot rate (or add negative points) - Forward rate is lower than spot - Foreign currency trades at a **forward discount**

Example with Japanese Yen:

```
# Spot rates for USD/JPY
spot_bid_jpy <- 118.27
spot_ask_jpy <- 118.37

# 3-month forward points (NEGATIVE)
forward_points_bid_jpy <- -143
forward_points_ask_jpy <- -140

# Calculate outright forward rates
# For JPY, points are in last two decimal places (hundredths)
forward_bid_jpy <- spot_bid_jpy + (forward_points_bid_jpy / 100)
forward_ask_jpy <- spot_ask_jpy + (forward_points_ask_jpy / 100)

cat("Given:\n")
```

Given:

```
cat(" USD/JPY Spot Bid:", spot_bid_jpy, "\n")
```

USD/JPY Spot Bid: 118.27

```
cat("  USD/JPY Spot Ask:", spot_ask_jpy, "\n")
```

USD/JPY Spot Ask: 118.37

```
cat("  3-month Forward Points:", forward_points_bid_jpy, "/", forward_points_ask_jpy, "\n\n")
```

3-month Forward Points: -143 / -140

```
cat("Outright Forward Calculation:\n")
```

Outright Forward Calculation:

```
cat("  Forward Bid =", spot_bid_jpy, "+", forward_points_bid_jpy, "/ 100\n")
```

Forward Bid = 118.27 + -143 / 100

```
cat("  Forward Bid =", forward_bid_jpy, "\n\n")
```

Forward Bid = 116.84

```
cat("  Forward Ask =", spot_ask_jpy, "+", forward_points_ask_jpy, "/ 100\n")
```

Forward Ask = 118.37 + -140 / 100

```
cat("  Forward Ask =", forward_ask_jpy, "\n\n")
```

Forward Ask = 116.97

```
cat("Result:\n")
```

Result:

```
cat("  USD/JPY 3-month Forward:", forward_bid_jpy, "/", forward_ask_jpy, "\n")
```

USD/JPY 3-month Forward: 116.84 / 116.97

```
cat("\nThe forward rate is LOWER than spot rate.\n")
```

The forward rate is LOWER than spot rate.

```
cat("The yen trades at a forward PREMIUM (more yen per dollar forward than spot).\n")
```

The yen trades at a forward PREMIUM (more yen per dollar forward than spot).

```
cat("Or equivalently, the dollar trades at a forward DISCOUNT against yen.\n")
```

Or equivalently, the dollar trades at a forward DISCOUNT against yen.

Forward Premium Formula:

The relationship between spot and forward rates, calculated as a percentage difference, is called the **forward premium**:

$$f = \frac{\text{Spot} - \text{Forward}}{\text{Forward}} \times \frac{360}{\text{days}} \times 100$$

Where f is the forward premium (or discount if negative), stated as an annual percentage.

Example Calculation:

```
# Using EUR/USD from above
spot_rate <- 1.0897
forward_rate_3m <- 1.0950
days <- 90 # 3 months = 90 days

# Calculate forward premium (annualized)
forward_premium <- ((spot_rate - forward_rate_3m) / forward_rate_3m) * (360 / days) * 100

cat("Forward Premium Calculation:\n")
```

Forward Premium Calculation:

```
cat(" Spot rate:", spot_rate, "\n")
```

Spot rate: 1.0897

```
cat(" 3-month forward rate:", forward_rate_3m, "\n")
```

3-month forward rate: 1.095

```
cat(" Days:", days, "\n\n")
```

Days: 90

```
cat("Formula: f = [(Spot - Forward) / Forward] × (360 / days) × 100\n\n")
```

Formula: $f = [(Spot - Forward) / Forward] \times (360 / days) \times 100$

```
cat(" f = [(", spot_rate, "-", forward_rate_3m, ") /", forward_rate_3m, "] × (360 /", days, ", "
```

$f = [(1.0897 - 1.095) / 1.095] \times (360 / 90) \times 100$

```
cat(" f =", round(forward_premium, 4), "%\n\n")
```

$f = -1.9361 \%$

```
if (forward_premium < 0) {  
  cat("The euro trades at a forward PREMIUM (negative premium = premium for base currency)\n")  
  cat("Or: The dollar trades at a forward DISCOUNT against the euro\n")  
} else {  
  cat("The euro trades at a forward DISCOUNT\n")  
  cat("Or: The dollar trades at a forward PREMIUM against the euro\n")  
}
```

The euro trades at a forward PREMIUM (negative premium = premium for base currency)
Or: The dollar trades at a forward DISCOUNT against the euro

Interpretation:

We'll explore forward premiums and discounts in much greater detail in Chapter 6 when we study Interest Rate Parity. The key insight:

Forward premiums/discounts are determined primarily by interest rate differentials between the two currencies.

Swap Rates:

Forward quotations of **two years or longer** are called **swap rates** rather than **cash rates**.

Why? Because they're primarily used for currency swaps (simultaneous borrowing/lending) rather than outright forward transactions.

Bottom Line on Forward Quotes:

- Start with spot rate (outright)
- Apply forward points (additions or subtractions)
- Result is outright forward rate
- Forward premium/discount reflects interest differentials (Chapter 6!)

Conclusion and Synthesis

Let me bring together the major themes from today's comprehensive look at the foreign exchange market.

3.1 Key Takeaways

1. The Foreign Exchange Market Serves Critical Functions

- **Transfer of purchasing power** across currencies and borders
- **Provision of credit** to finance international trade's time lags
- **Risk management** through hedging instruments
- **Price discovery** through continuous global trading

Without these functions, modern international commerce would be impossible.

2. Market Structure Has Been Revolutionized

We've witnessed an extraordinary transformation:

- From telephone-based voice trading to algorithm-dominated electronic execution
- From a closed, bank-only market to one accessible to participants of all sizes
- From high settlement risk to near-elimination through CLS
- From limited transparency to real-time global price information

Yet the fundamental players remain: dealers, commercial transactors, speculators, central banks, and brokers.

3. Transaction Types Serve Different Needs

- **Spot:** Immediate needs (T+2 settlement)
- **Forwards:** Locking in future rates, hedging future exposures
- **Swaps:** Temporary currency exchanges, managing liquidity
- **NDFs:** Access to restricted currencies

Each has its place in a comprehensive FX risk management strategy.

4. Size and Scope Are Staggering

- **\$6.6 trillion per day** in trading volume
- London and New York dominate but Asian centers growing
- U.S. dollar involved in 88% of transactions
- Emerging market currencies rapidly increasing their share

This is truly a global, 24-hour market.

5. Quotation Conventions Matter

Understanding how to read and interpret currency quotations is fundamental:

- Base currency vs. price currency
- European terms vs. American terms

- Direct vs. indirect quotes
- Bid vs. ask rates
- Cross rates and arbitrage
- Forward points and premiums

Mastery of these concepts is essential for anyone working in international finance.

3.2 Connecting to Broader Course Themes

From Chapter 1: The foreign exchange market is a key component of the global financial marketplace we studied. It provides the linkages between different currency zones.

Looking Ahead to Chapter 6: We'll study **International Parity Conditions**—the economic relationships that determine forward premiums/discounts and explain why forward rates differ from spot rates. Interest Rate Parity, Purchasing Power Parity, and the Fisher Effect will complete our understanding of currency pricing.

Looking Further Ahead:

- **Chapter 7:** Currency derivatives (options and futures) that trade in this market
- **Chapter 8:** Interest rate swaps and currency swaps
- **Later chapters:** How firms use these instruments for hedging, speculation, and managing international operations

3.3 Practical Applications

For Corporate Treasurers:

- Understanding market structure helps in choosing execution venues
- Knowing bid-ask spreads informs hedging decisions
- Cross-rate calculations are essential for multi-currency operations

For Financial Analysts:

- Currency movements impact international investment returns
- Forward rates provide information about market expectations
- Arbitrage relationships reveal market efficiency

For Economists:

- FX market activity reflects global capital flows
- Central bank intervention reveals policy priorities
- Currency crises often originate in FX market dynamics

For All of Us:

- Exchange rates affect prices of imported goods
- Currency movements impact travel costs
- Understanding FX helps make sense of global economic news

3.4 The Mini-Case: Iceland

Your textbook's mini-case on Iceland provides a sobering real-world example of what happens when a small country's currency comes under severe pressure. The case illustrates:

- How capital flows can overwhelm small economies
- The challenges of maintaining independent monetary policy
- The relationship between interest rates and currency values
- The difficulty of defending a currency against market forces
- The importance of adequate foreign exchange reserves

We'll discuss this case in detail to reinforce today's concepts.

3.5 Preparing for Success

To master this material:

1. **Practice reading quotations:** Look at FX quotes in the Wall Street Journal or on Bloomberg. Identify base/quote currencies.
2. **Calculate cross rates:** Given USD/EUR and USD/JPY, calculate EUR/JPY
3. **Work arbitrage problems:** Practice triangular arbitrage calculations
4. **Understand bid-ask:** Always think about which side of the market you're on
5. **Follow the market:** Track a few currency pairs daily—notice how they move

Resources:

- Bloomberg: Real-time FX data
- Financial Times: Daily FX commentary
- Federal Reserve: Historical exchange rate data
- BIS: Triennial Central Bank Survey (comprehensive market data)
- WSJ: Daily closing rates

3.6 Common Pitfalls to Avoid

1. Confusing Base and Quote Currency

Always remember: currency on LEFT of slash is base (always one unit).

2. Mixing Up Bid and Ask

You buy at dealer's ask, sell at dealer's bid. The spread is their profit.

3. Cross Rate Errors

Pay careful attention to which currency is in numerator and denominator.

4. Forward Points Mistakes

Remember to divide points by appropriate amount (10,000 for EUR, 100 for JPY) and watch for sign (positive or negative).

5. Percentage Change Direction

When spot changes from 1.20 to 1.30, that's appreciation of foreign currency (takes more dollars per unit), NOT dollar appreciation.

3.7 Questions for Reflection

As you review this chapter, consider:

1. How do bid-ask spreads in FX markets compare to those in stock markets? Why might they differ?
2. Why has electronic trading grown so dramatically? What are the benefits and risks?
3. How does the 24-hour nature of the FX market create both opportunities and challenges?
4. What role should central banks play in FX markets? When is intervention justified?
5. How might blockchain technology further transform FX markets in the future?

3.8 Looking Ahead

Next Session:

In Chapter 6, we'll study **International Parity Conditions**—the fundamental economic relationships that link:

- Exchange rates
- Interest rates
- Inflation rates
- Forward rates

These parity conditions explain WHY currencies move as they do and WHY forward rates differ from spot rates. We'll derive Interest Rate Parity, study Purchasing Power Parity, and explore whether these relationships hold in practice.

Assignment:

- Read Chapter 6 thoroughly
- Work through the end-of-chapter problems for Chapter 5
- Find a recent article about a currency crisis or significant FX movement
- Be prepared to explain the movement using concepts from this chapter

3.9 Final Thoughts

The foreign exchange market is where theory meets practice in international finance. Every concept we've studied today—market structure, transaction types, quotation conventions—has direct application in the real world.

You're now equipped to:

- Read and interpret FX quotations in any financial publication
- Understand how the global FX market operates
- Calculate cross rates and identify arbitrage opportunities
- Appreciate the revolution in FX trading technology
- Recognize different transaction types and their uses

This knowledge forms the foundation for everything else we'll study in international finance. Foreign exchange is the medium through which all international financial transactions occur. Master these concepts, and you've taken a major step toward understanding global finance.

Thank you for your attention today. Please don't hesitate to reach out with questions as you work through the material.

See you next class!

Questions and Discussion

Check for Understanding:

1. "Can someone explain in their own words what makes a swap transaction different from an outright forward?"
2. "If you see EUR/USD 1.2174, which currency is more expensive? How many dollars does one euro cost?"
3. "What's the difference between European terms and American terms?"
4. "Why might a forward rate be different from a spot rate? We'll explore this in detail next chapter, but what's your intuition?"
5. "How has technology transformed the FX market? What are the benefits and risks?"

Application Questions:

1. "If you're a U.S. company expecting to receive €10 million in 6 months, how could you use the forward market to hedge your risk?"
2. "Given these three rates, is there an arbitrage opportunity? How would you exploit it?"
3. "Why would a central bank intervene in the FX market? When might intervention fail?"